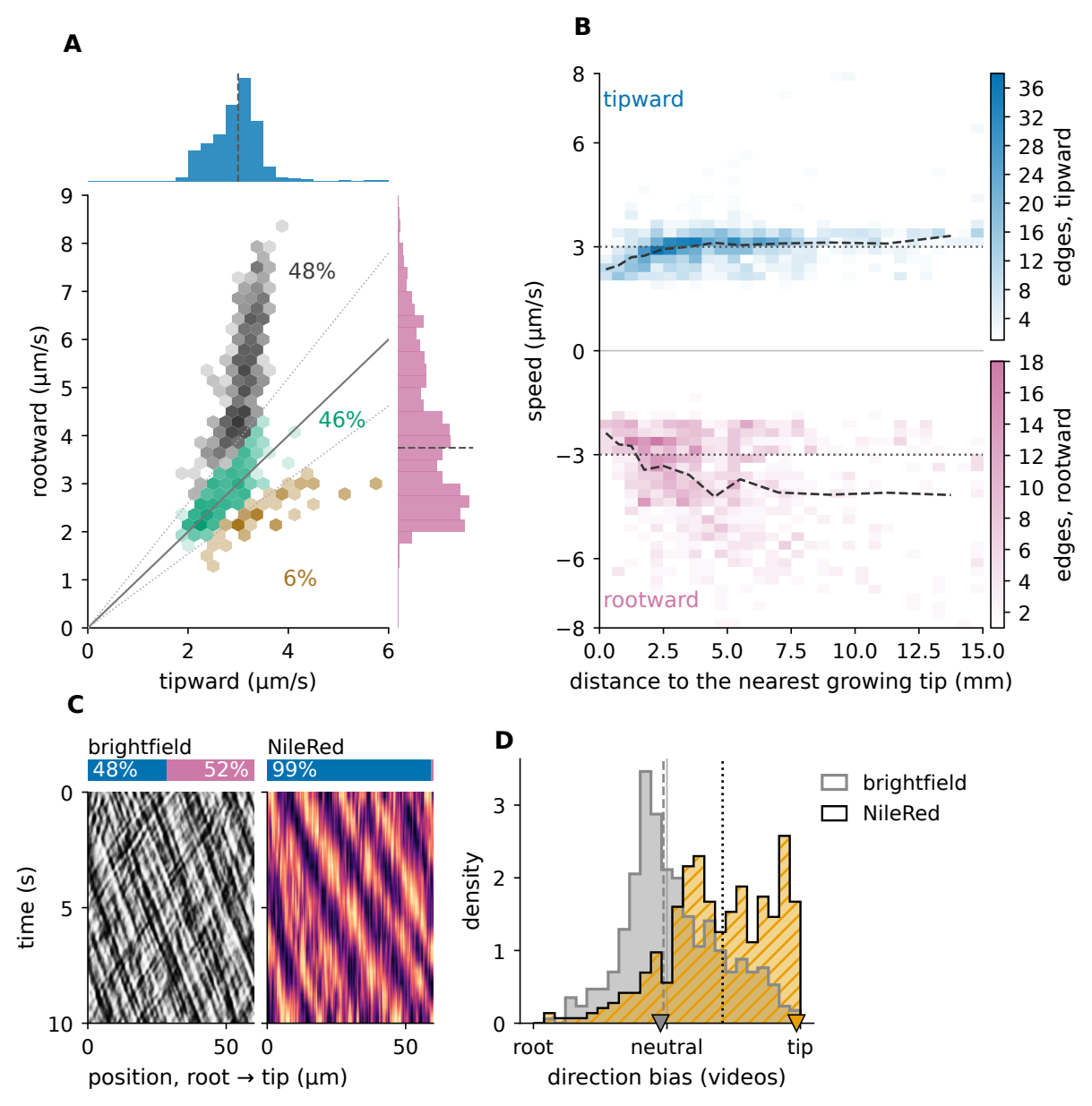

A
B
C
D
rootward (µm/s)
tipward (µm/s)
48%
46%
6%
tipward
rootward
speed (µm/s)
edges, tipward
edges, rootward
distance to the nearest growing tip (mm)
brightfield
NileRed
48%
52%
99%
time (s)
position, root → tip (µm)
density
brightfield
NileRed
root
neutral
tip
direction bias (videos)